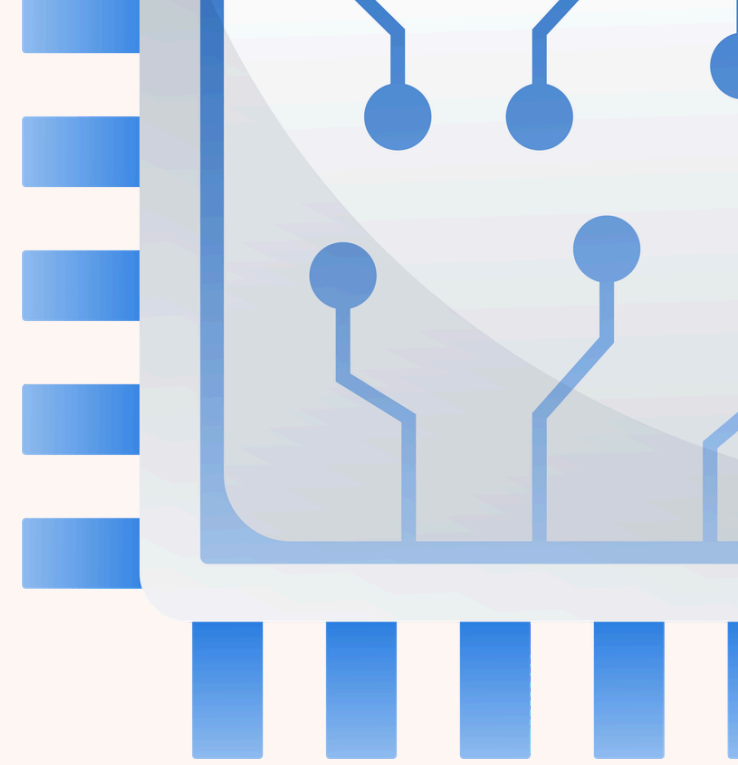
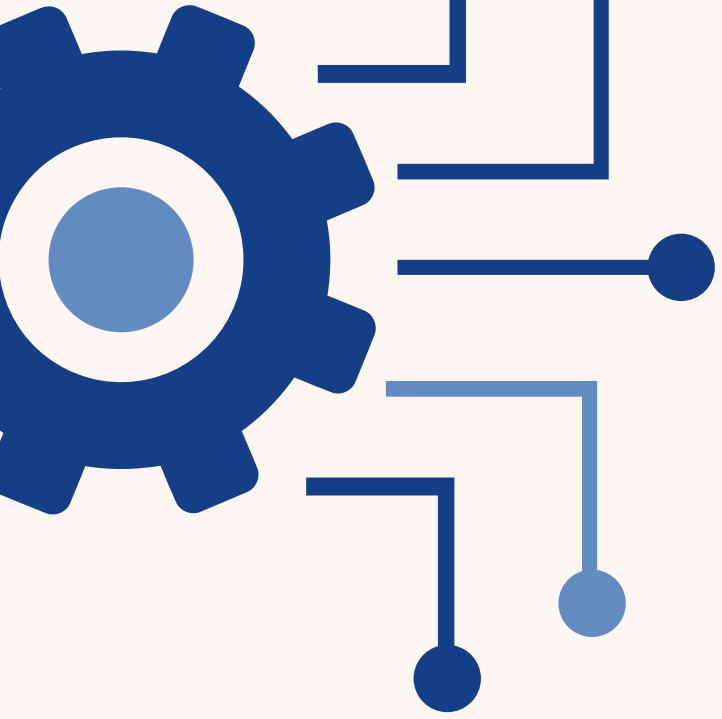
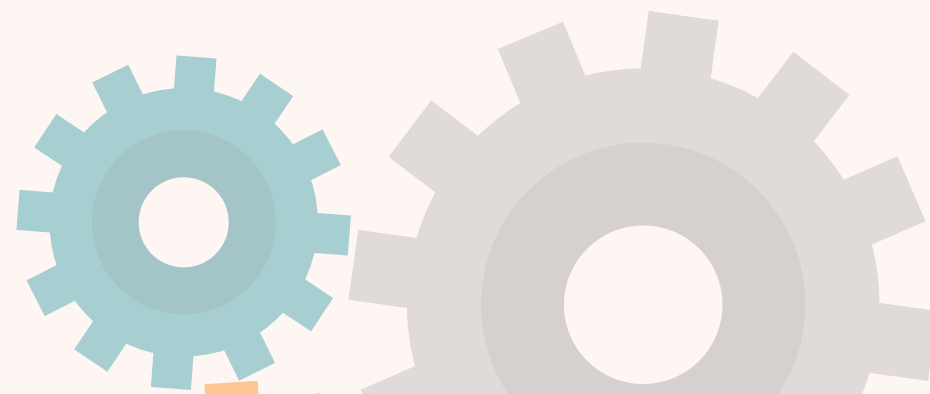
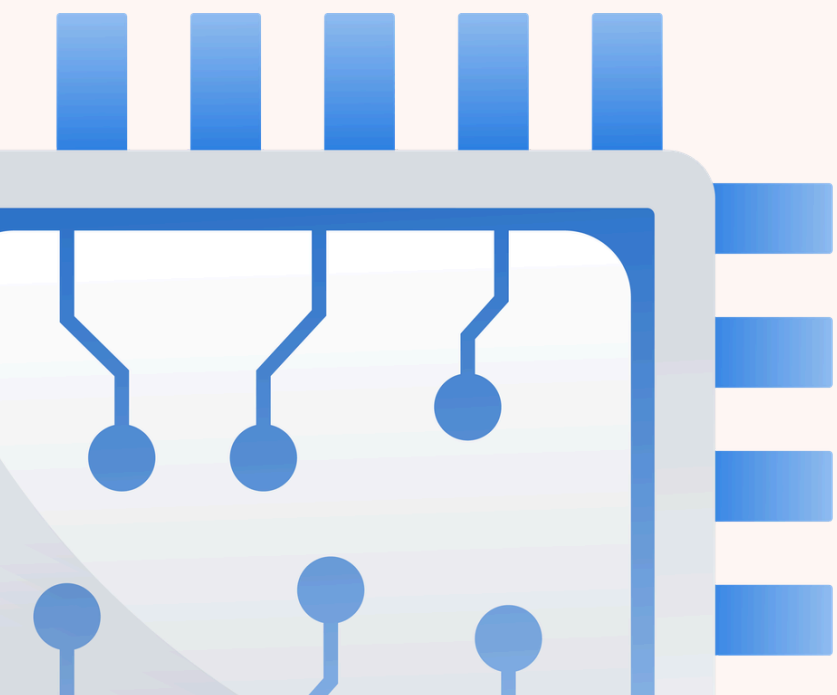




STEERING CAR

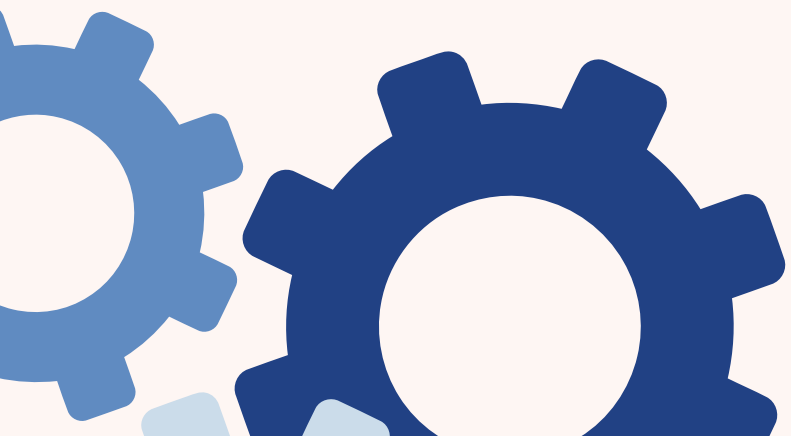
Presentation



PROJECT OVERVIEW

Building an intelligent steering car that combines steering precision, dynamic speed control, and autonomous safety features.

Designed with real-time hardware responsiveness and no-lag user interrupts.



HARDWARE ARCHITECTURE

Brain & Power:

+ Arduino Uno
+ L298N Dual
Motor Driver.

Motors :

2x DC Motors
+ Servo Motor
(Steering via
Potentiometer).

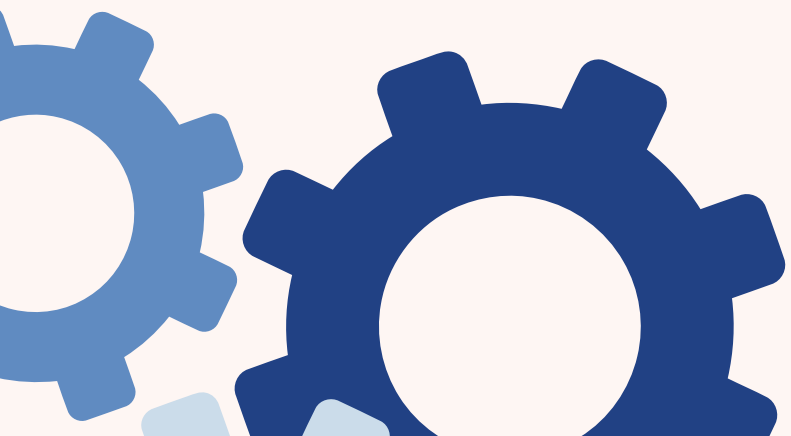
Sensors & Display :

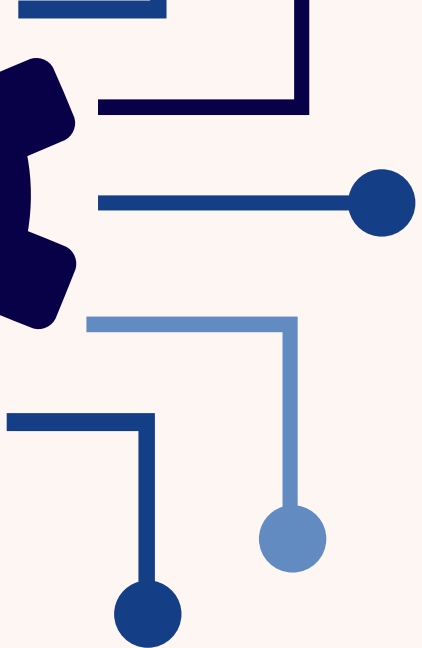
2x Ultrasonic
Sensor (HC-SR04)
+ 16x2 I2C LCD
Display .
+2x potentiometer
+2x pushbutton



FAILSAFE & SAFETY SYSTEM

- Active Protection: Dual ultrasonic sensors continuously scan the path.
- Collision Avoidance: Automatic emergency stop triggered if any obstacle is detected within 20 cm.
- Priority: Safety execution overrides standard motion commands inside the logic loop.





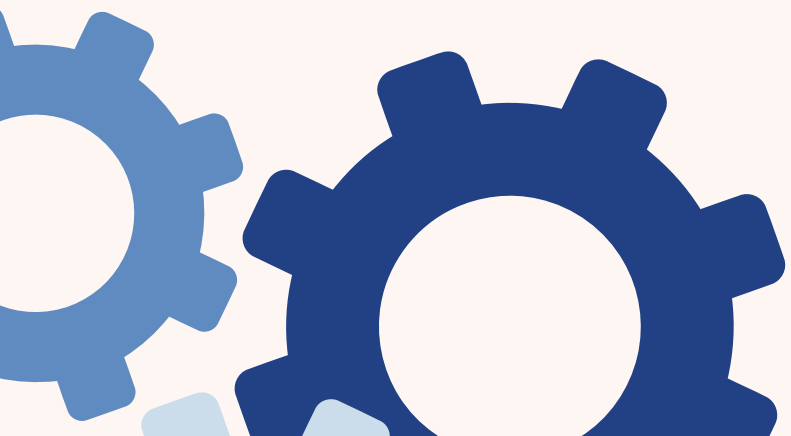
ADVANCED FEATURES



Hardware Interrupt (Pin 2): Uses `attachInterrupt()` for instant direction toggling.

Software Debouncing: Non-blocking debouncing to prevent false triggers.

Why it matters? Ensures immediate responsiveness without lagging the main program loop.



LIVE DATA

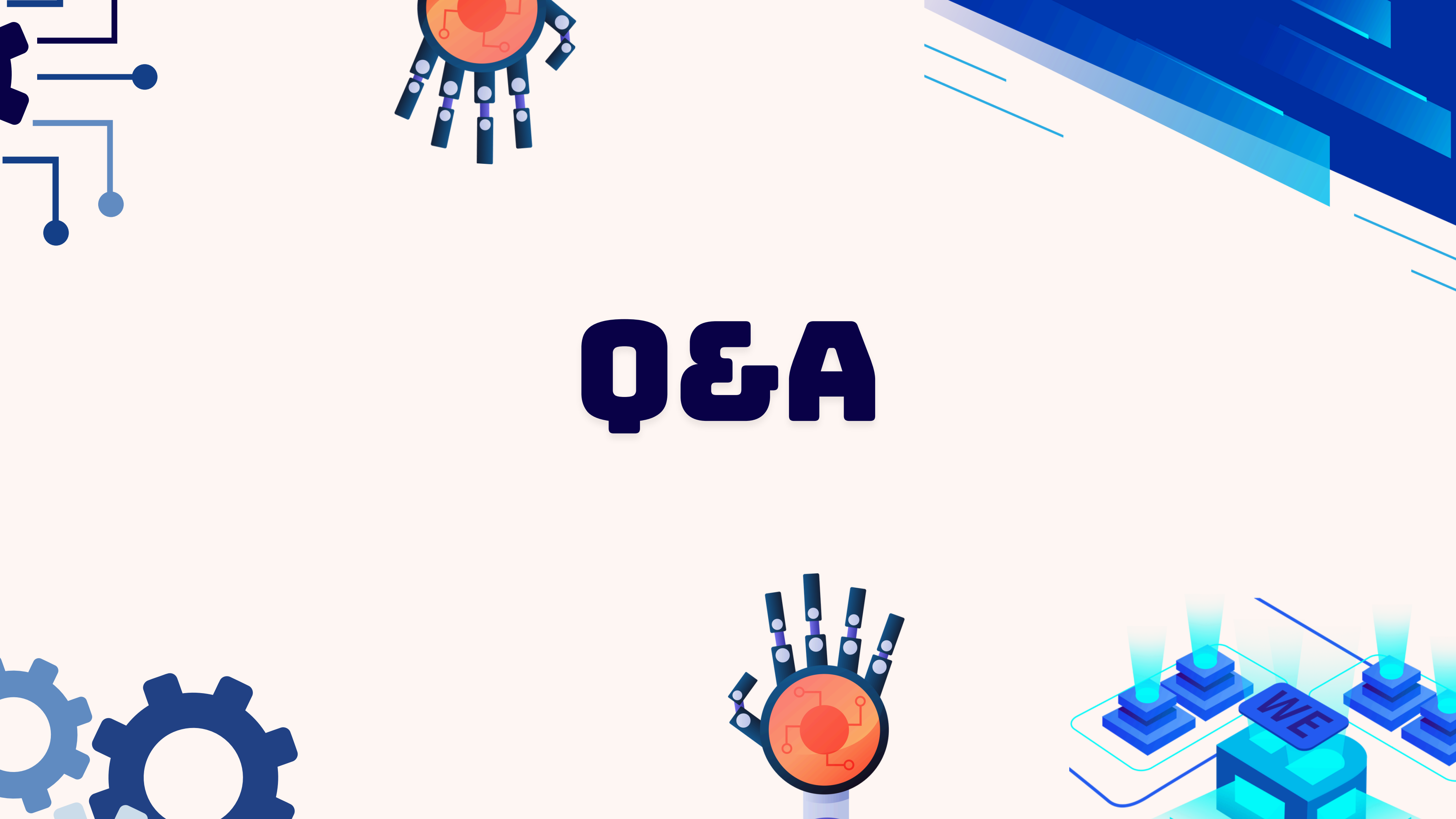
- **Live Calculations:** Real-time speed (m/s) and RPM computed on the fly.
- **On-Demand Display:** Required calculations is outputted to the LCD and Serial Monitor upon button press.
- **Resource Efficiency:** Uses I2C protocol to save Arduino pins for other peripherals.

CONCLUSION

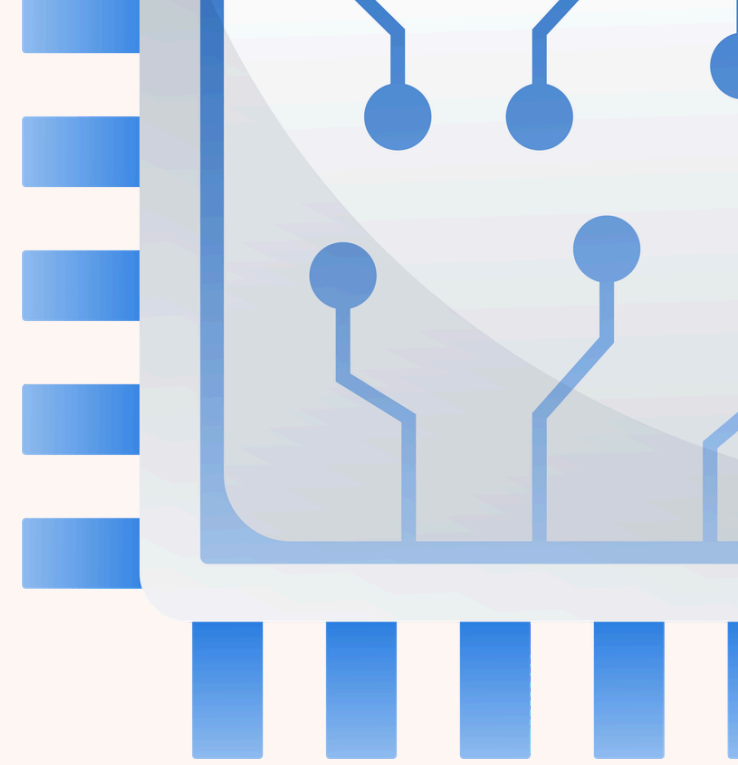
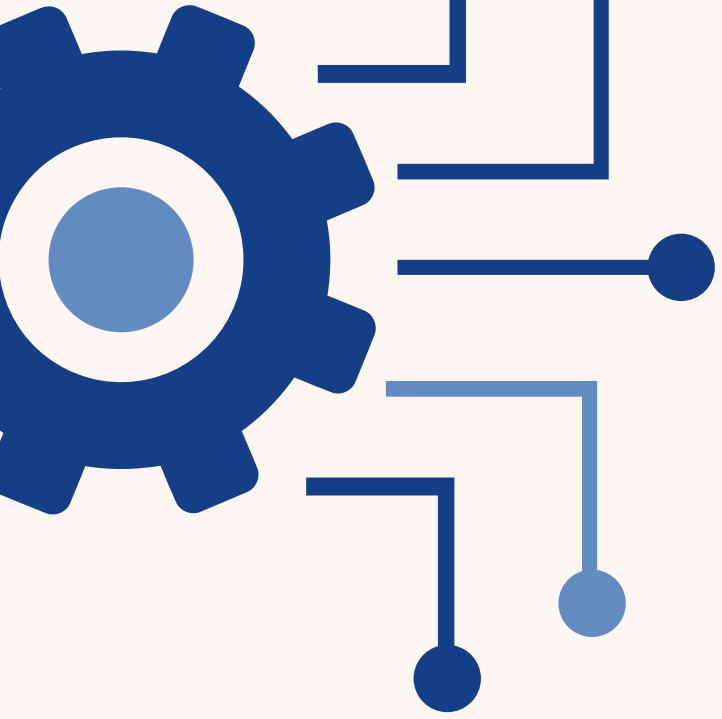
Successful integration of Hardware Interrupts, Failsafe logic, and Telemetry.

Live simulator on Tinkercad :

<https://www.tinkercad.com/things/lkNFYM02Fw9-martinn-final-steering-car?sharecode=eKNPkrG-7DpUVj-WDK0rt8aDHa1EXqgV8EGHZmQ-iPM>



Q&A



**THANK
TOU**

